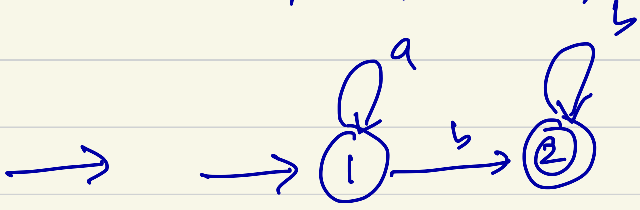
Week 4, Lecture 1

Feb 12, 2024

There are languages that are not regular.

$$L = \{a^n b^n \mid n \geq 0,\ a, b \in \Sigma\} \qquad \Sigma = \{a, b\}$$

$$L = \{\varepsilon, ab, aabb, aaabbb, \ldots\}$$



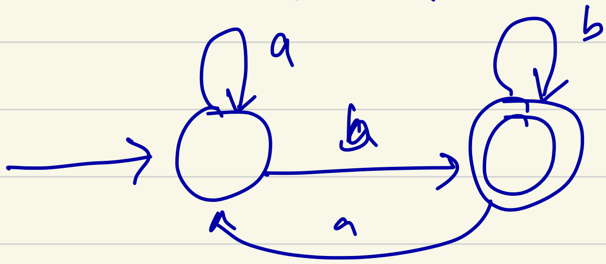

```
bool    p(int n) {
        // returns true if n is odd
        // return false if n is even
}

bool    prime(int n) {
        // returns true if n is prime
        // false otherwise
}
```

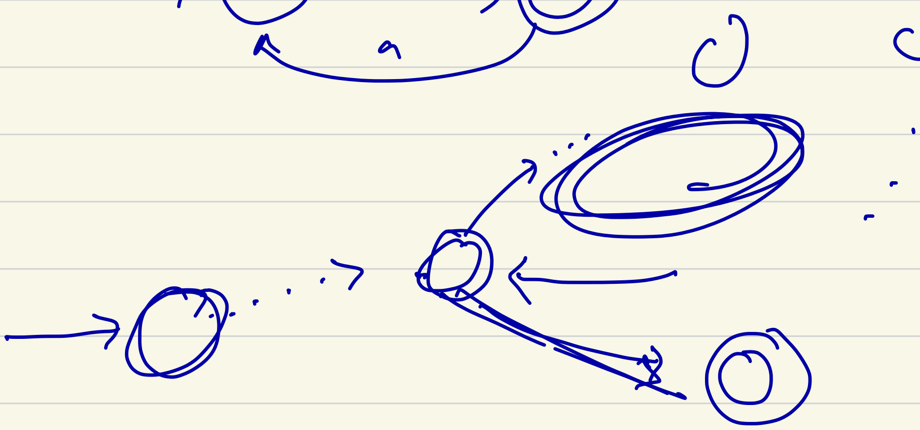
Pumping Lemma:

if A is a RL, then there is a number p (the pumping length) where if s is any string in A of length at least p, then s may be divided into three pieces $s = xyz$, satisfying following conditions.

1. $\forall i \geq 0$, $xy^iz \in A$
2. $|y| > 0$
3. $|xy| \leq p$

Proof idea: Imagine we have a regular language (infinite)





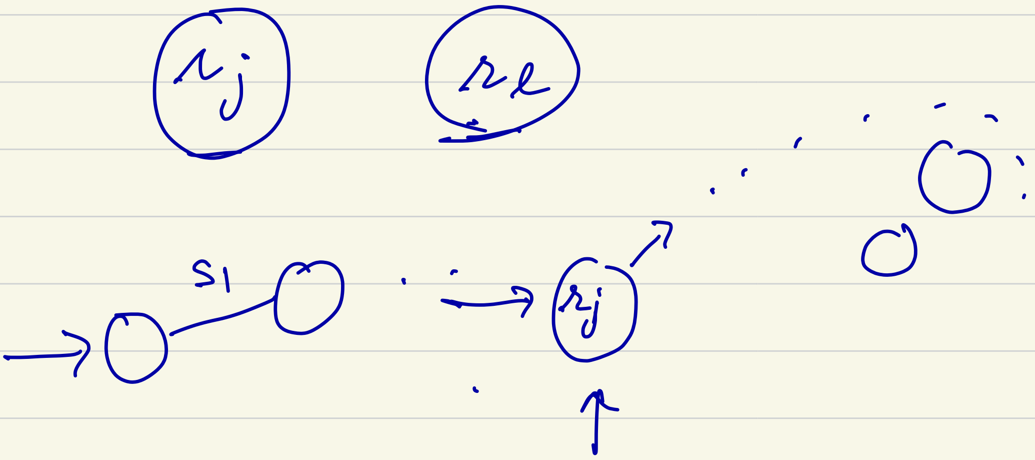
Proof: $M = (Q, \Sigma, \delta, q_1, F)$ be the DFA for A.

$S = s_1 s_2 \ldots s_n$ $\quad |s| = n$, $n \geq p$. $\quad p = |Q|$

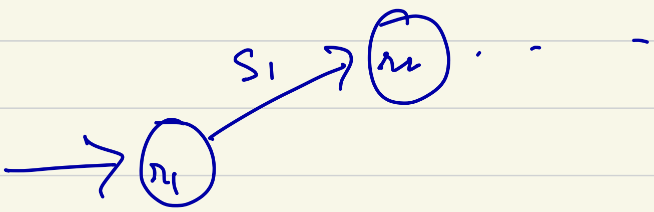
Let $\boxed{r_1, r_2, \ldots, r_{n+1}}$ be the states that machine M enters when s is passed to M.

$$r_{i+1} = \delta(r_i, s_i) \quad \forall i \quad 1 \leq i \leq n$$

$r_1, r_2, \ldots, r_{n+1}$ length of this sequence is $n+1$.



r_l must occur among the first $p+1$ places.



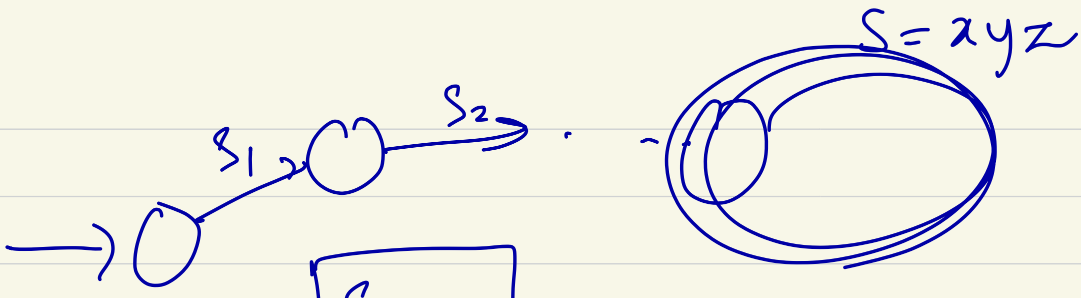
$r_1, r_2, \ldots r_j, \ldots r_l, \ldots, r_{n+1}$

$\boxed{l \leq p+1}$ $S = s_1 s_2 \ldots s_n$

Let $x = s_1 s_2 \ldots s_{j-1}$

$y = s_j s_{j+1} \ldots s_{l-1}$

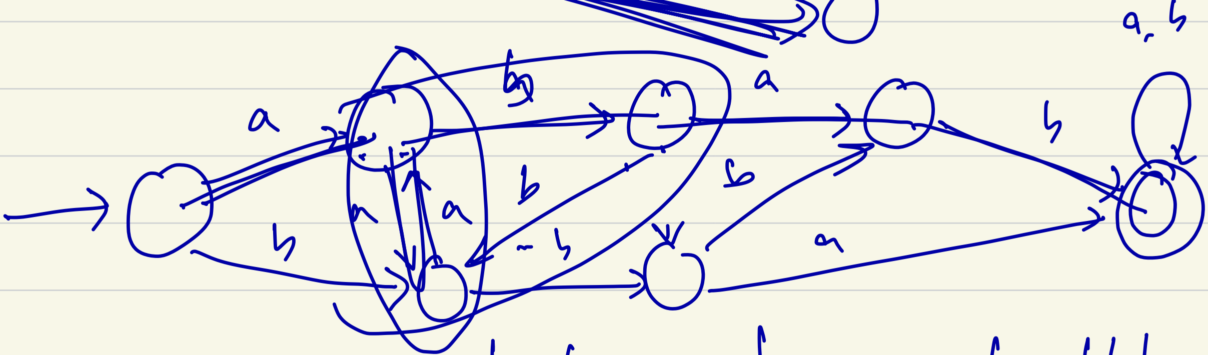
$z = s_l \ldots s_n$



$\boxed{S_{j-1}}$

$\boxed{s_1, s_2, \ldots, s_{j-1} = x}$

$$\underbrace{s_1\, s_2\, s_3 \ldots s_{j-1}}_{x}\ \underbrace{\boxed{s_j\, s_{j+1} \ldots s_{l-1}}}_{y}\ \underbrace{s_l\, s_{l+1} \ldots s_n}_{z}$$



input a · bab | aaa ba bbb

a aa b a b

$y = aa$

x takes M from r_1 to r_j

y .. .. r_j to r_l

z .. r_l to r_{n+1}

so if $xyz \in A$, $xy^i z \in A$

$(0 \cup 1)^* 11 (0 \cup 1)^*$

$\underline{A}$ $\quad$ $\underline{PL}$

1. A is regular
2. $s \in A$ s.t. $|s| \geq p$
3. $s = xyz$
4. $\forall i \geq 0 \quad xy^iz \in A$
5. $|y| > 0$
6. $|xy| \leq p$

$A = \{ww \mid w \in \Sigma^*\}$ $\qquad \Sigma = \{0,1\}$

prove A is not regular.

Proof: Suppose A is regular. Let p be the pumping length.

$s \in A \qquad |s| \geq p$

$s = 0^p 1 0^p 1 \qquad |s| = 2p+2 \geq p$

$s = xyz$

1) $\forall i \geq 0 \quad xy^iz \in A$

✓ 2) $|y| > 0$

✓ 3) $|xy| \leq p$

$x = 0^{p-1}, \quad y = 0, \quad z = 10^p1$

$\underbrace{0^{p-1} \cdot 0} = p$ copies of 0

$i = 1 \quad xyz \in A, \quad i = 2, \; i = 3, \; i = 4 \ldots$

$i = 0$

$$0^{p-1}(0)^2 1 0^p 1$$

$$= 0^{p-1} \cdot 0^2 \; 1 \; 0^p \; 1$$

$$= \underline{0^{p+1}} \; 1 \; \underline{0^p} \; 1 \notin A$$

A is not regular.